

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)**UNIT TEST****09-08-2020****JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]****PAPER-1****Time : 3 Hours****Maximum Marks : 180****READ THE INSTRUCTIONS CAREFULLY****GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

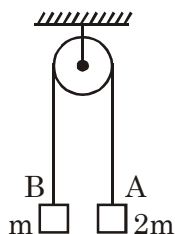
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

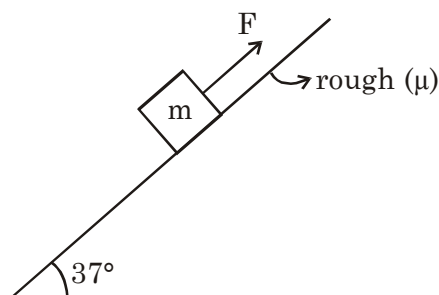
1. In given diagram pulley and string are ideal, block A of mass $2m$ and B of mass m are released from rest then :



- (A) Magnitude of acceleration of A with respect to B is $\frac{2g}{3}$.
- (B) Displacement of A & B will be equal.
- (C) Distance covered by A & B in 3s will be 15 m.
- (D) Velocity of A and B are equal.

Space for Rough Work

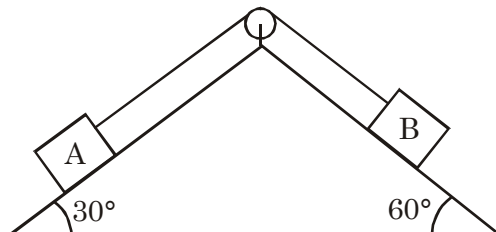
2. Choose the **CORRECT** statement(s) :
- (A) Force depends on frame of reference.
 (B) Relative velocity of two objects is independent of frame of reference.
 (C) In uniform circular motion velocity of object is variable.
 (D) In uniform circular motion acceleration of object remains constant.
3. A particle is moving on a circle in horizontal plane. At any instant its speed is 4 m/s and acceleration is at angle 30° from velocity vector. Radius of circle is 2m. Then :
- (A) Acceleration of particle is 16 m/s^2 .
 (B) Speed is changing at a rate of $8\sqrt{3} \text{ m/s}^2$.
 (C) $\vec{a} \cdot \vec{v} = 32\sqrt{3}$
 (D) $|\vec{a} \times \vec{v}| = 32$
4. In given diagram $m = 4\text{kg}$, $\mu = 0.5$. Force F is applied on block. Select the **CORRECT** statement(s). In all case block is not moving :- ($g = 10 \text{ m/s}^2$)



- (A) If $F = 24 \text{ N}$ then friction force on the block is zero.
 (B) If $F = 10 \text{ N}$ then friction force acting on the block is 14 N .
 (C) If $F = 16 \text{ N}$ then friction force acting the block is 18 N .
 (D) If $F = 12 \text{ N}$ then friction on the block is 12 N .

Space for Rough Work

5. Two blocks A and B of masses 10 kg and 12 kg respectively are kept on a rough wedge as shown in figure. The coefficient of friction between the block A and wedge is 0.6, while that between the wedge and B is 0.3. The blocks are connected by a light inextensible thread. The wedge is fixed with respect to ground. ($g = 10 \text{ m/s}^2$)



- (A) Acceleration of block A is $\left(\frac{6.8\sqrt{3}-5}{22}\right)g$ down the plane.
- (B) Acceleration of block A is $\left(\frac{6\sqrt{3}-5}{22}\right)g$ up the plane.
- (C) Tension in the string connecting A and B is 55.79 N.
- (D) Friction force between block A and incline is 35.92 N.
6. A car runs around a curve of radius 10 m at a constant speed of 10 ms^{-1} . Consider the time interval for which car covers a curve of 120° arc :-
- (A) Resultant change in velocity of car is $10\sqrt{3} \text{ ms}^{-1}$
- (B) Instantaneous acceleration of car is 10 ms^{-2}
- (C) Average acceleration of car is $\frac{5}{24} \text{ ms}^{-2}$
- (D) Instantaneous and average acceleration are same for the given period of motion.

Space for Rough Work

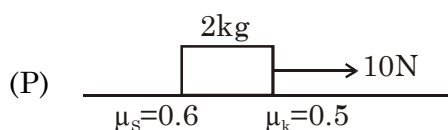
SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

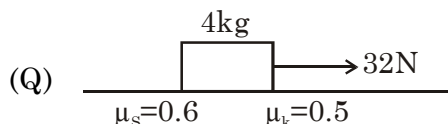
7. In list-I blocks are initially at rest on rough horizontal floor, and forces are applied on the blocks at $t = 0$ as shown. List-II represents friction, acceleration and speed of blocks. Match the **CORRECT** option(s) of both list : ($g = 10 \text{ m/s}^2$)

List-I

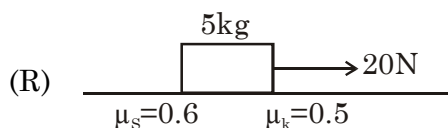
List-II



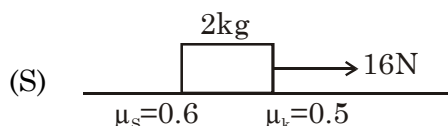
- (1) Friction acting on block is 20 N



- (2) Acceleration of block is zero



- (3) Speed of block after 3 sec is 9 m/s



- (4) Friction acting on block is 10 N

- (5) Acceleration of block is 3 m/s^2

(A) $P \rightarrow 2, 4$; $Q \rightarrow 1, 3, 5$; $R \rightarrow 1, 2$; $S \rightarrow 3, 4, 5$

(B) $P \rightarrow 2, 4$; $Q \rightarrow 2, 3, 5$; $R \rightarrow 1, 3$; $S \rightarrow 1, 2, 3$

(C) $P \rightarrow 1, 5$; $Q \rightarrow 1, 3, 5$; $R \rightarrow 2, 4$; $S \rightarrow 2, 4$

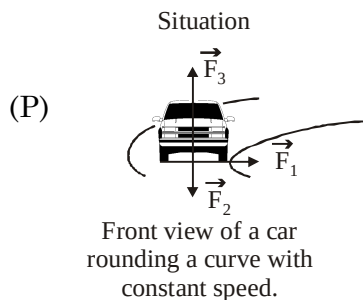
(D) $P \rightarrow 3, 4$; $Q \rightarrow 3, 4, 5$; $R \rightarrow 4, 5$; $S \rightarrow 2, 3, 5$

Space for Rough Work

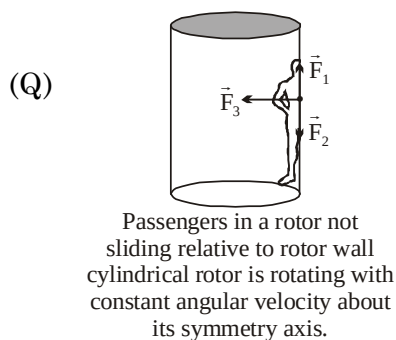
8. List-I shows certain situations and list-2 shows information about forces.

List-I

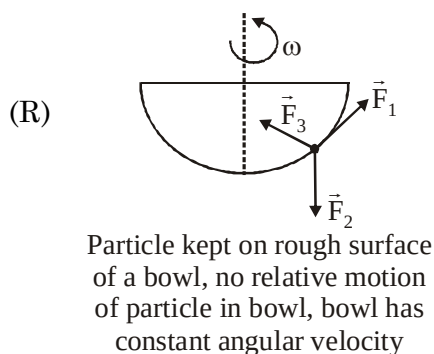
List-II



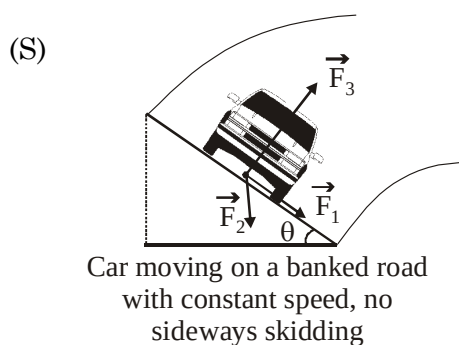
(1) $\vec{F}_1 + \vec{F}_2 + \vec{F}_3$ is centripetal force.



(2) $\vec{F}_1$ is static friction.



(3) $\vec{F}_1$ can be in direction opposite to that shown in figure.



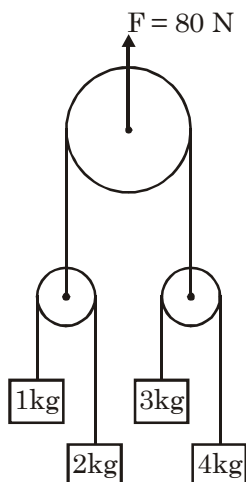
(4) $\vec{F}_1 + \vec{F}_2 = \vec{0}$

(5) $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}$

- (A) P $\rightarrow$ 2,4; Q $\rightarrow$ 2,3,4; R $\rightarrow$ 2,3; S $\rightarrow$ 1,2,3
 (B) P $\rightarrow$ 1,2; Q $\rightarrow$ 1,2,4; R $\rightarrow$ 1,2,3; S $\rightarrow$ 1,2,3
 (C) P $\rightarrow$ 3,5; Q $\rightarrow$ 3,4,5; R $\rightarrow$ 4,5; S $\rightarrow$ 2,3,5
 (D) P $\rightarrow$ 1,2; Q $\rightarrow$ 1,2,5; R $\rightarrow$ 2,3; S $\rightarrow$ 3,4

Space for Rough Work

9. In the figure shown, all the pulleys are massless and strings are light ($g = 10 \text{ m/s}^2$)



List-I (Mass of blocks)

- (P) 1 kg block
(Q) 2 kg block
(R) 3 kg block
(S) 4 kg block

List-II (Motion of Blocks)

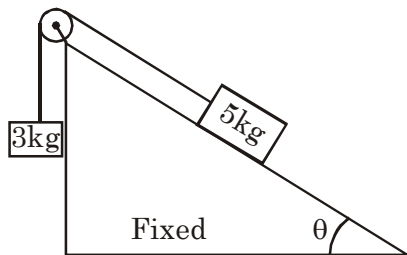
- (1) Will remain stationary
(2) will move down
(3) will move up
(4) 5 m/s^2
(5) 10 m/s^2

- (A) $P \rightarrow 1$; $Q \rightarrow 3,5$; $R \rightarrow 2$; $S \rightarrow 1,4$
(C) $P \rightarrow 1,4$; $Q \rightarrow 3$; $R \rightarrow 2,4$; $S \rightarrow 1$

- (B) $P \rightarrow 3,5$; $Q \rightarrow 1$; $R \rightarrow 2$; $S \rightarrow 2,4$
(D) $P \rightarrow 2$; $Q \rightarrow 3,4$; $R \rightarrow 1,4$; $S \rightarrow 1,2$

Space for Rough Work

10. A system is shown in which angle of wedge can be changed. from 0 to 90°. (All surface are smooth, take angle in degree) ($g = 10 \text{ m/s}^2$)



List-I

- (P) Values of θ for which acceleration of 3 kg is upward
 (Q) Acceleration of 3 kg is $g/8$ in upward direction, for this θ is
 (R) Acceleration of 5 kg is $g/6$ up the inclined than θ is
 (S) Normal between inclined and 5 kg can be (in N)

List-II

- (1) 10
 (2) 20
 (3) 30
 (4) 53
 (5) 70

(A) $P \rightarrow 1,3$; $Q \rightarrow 1$; $R \rightarrow 4$; $S \rightarrow 3,4,5$

(B) $P \rightarrow 2,4$; $Q \rightarrow 2$; $R \rightarrow 2$; $S \rightarrow 1,3,4$

(C) $P \rightarrow 3,5$; $Q \rightarrow 3$; $R \rightarrow 1$; $S \rightarrow 2,3,4$

(D) $P \rightarrow 4,5$; $Q \rightarrow 4$; $R \rightarrow 3$; $S \rightarrow 1,2,3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

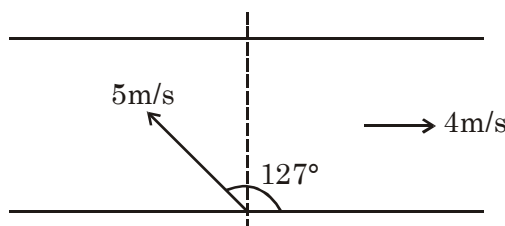
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

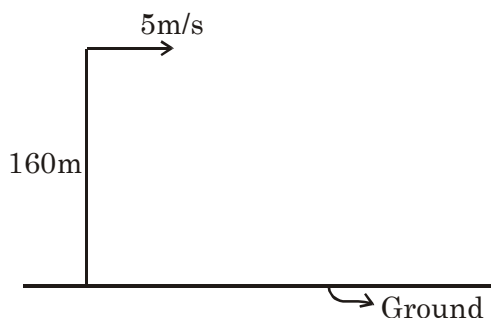
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. In the given figure river water is flowing at 4 m/s. A boat is steered at angle 127° from the direction of flow of river. Width of river is 1560 m, and velocity of boat in still water is 5 m/s. Find drift of boat in river in (km).

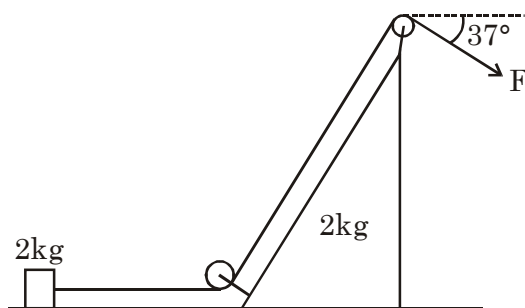


Space for Rough Work

2. A ball is projected horizontally at 5 m/s from the top of a tower of height 160 m. Find horizontal distance (in m) covered by the ball when it strike on horizontal ground (in m).
(Take : $g = 10 \text{ m/s}^2$)

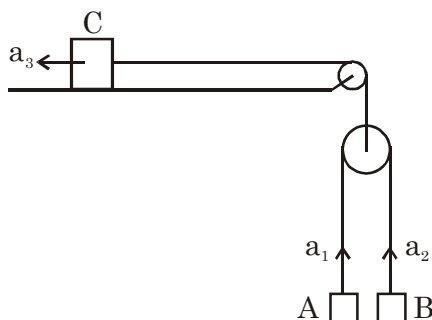


3. A small block of mass 2 kg and a wedge of mass 2 kg are placed on a smooth horizontal surface. Now string is pulled by applying force F of magnitude 8 N as shown in diagram pulley and string are ideal. Find acceleration (in m/s^2) of wedge.

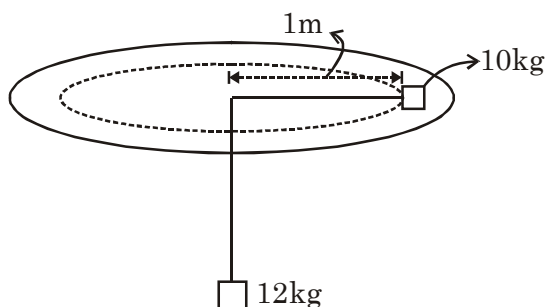


Space for Rough Work

4. In the given diagram pulley and strings are ideal $a_1 = 2 \text{ m/s}^2$, $a_2 = 3 \text{ m/s}^2$ mass of block C = 3kg. Find net force (in N) on block C.

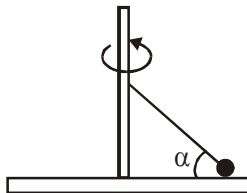


5. In a uniform gravitational field g , which is acting vertically downward, a ball is thrown at an angle with horizontal such that the initial (immediately after projection) radius of curvature is 8 times the minimum radius of curvature. If angle of projection is 150° , then find the value of θ (in degree). ($g = 10 \text{ m/s}^2$)
6. A mass 10 kg rotating freely in a horizontal circle of radius 1m on a frictionless table supports a stationary mass 12 kg, attached to the other end of the string passing through small hole O in table hanging vertically. Find angular velocity (in rad/sec) of rotation. ($g = 10 \text{ m/s}^2$)

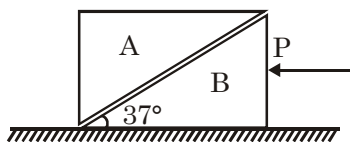


Space for Rough Work

7. A circular platform rotates around a vertical axis with angular velocity $\omega = 10 \text{ rad/s}$. On the platform is a ball of mass 100 gram, attached to the long axis of the platform by a thin rod of length 12 cm ($\alpha = 30^\circ$). Find normal force exerted by the ball on the platform (in newton). Friction is absent. ($g = 10 \text{ m/s}^2$)



8. Blocks A and B each have same mass $m = 1 \text{ kg}$. If the largest horizontal force P (in Newton) which can be applied to B so that A will not slip up on B, find $P/2$. Neglect any friction. ($g = 10 \text{ m/s}^2$)



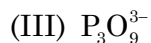
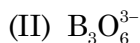
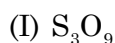
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following species has/have same number of X–O–X linkage where X = S or P or B ?



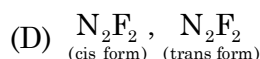
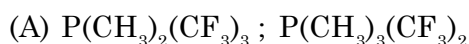
- (A) II & III

- (B) I, II & III

- (C) I & III

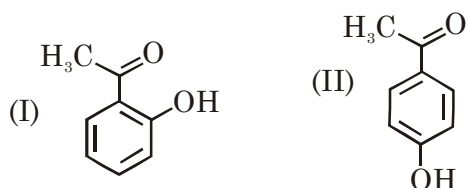
- (D) III & IV

- 2.** In which pair first compound has more dipole moment than second compound.



Space for Rough Work

3. Consider two isomeric compounds

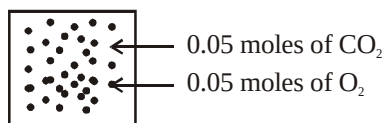


Select **INCORRECT** statement.

- (A) I have intermolecular H-bonding
 (B) II have intermolecular H-bonding
 (C) Boiling point of II < I
 (D) I is more volatile as compared to II
4. Which of the following is/are NOT correct -
 (A) 10 gm of N_2O and 10 gm of CO_2 contains equal number of atoms.
 (B) 14 gm of CO contains equal number of atoms present in 14 gm of C_2H_4 .
 (C) 4 moles of SO_2 and 3 moles of SO_3 contains equal number of atoms.
 (D) Equal masses of O_2 and O_3 contains equal number of molecules
5. 110 gm sample of P_4O_6 contains (Atomic weight of P = 31)
 (A) $3N_A$ number of O atoms
 (B) $5N_A$ number of total atoms
 (C) 62 gram of P atoms
 (D) $12N_A$ number of P – O bonds

Space for Rough Work

6. Which of the following is/are correct for the given closed container containing CO_2 & O_2 gas -



- (A) Moles of oxygen atom in container is 0.1
- (B) Mass of oxygen atoms in container is 3.2 gm
- (C) Total number of moles of all atoms in container is 0.25
- (D) Mass of carbon present in container is equal to the mass of 3.011×10^{22} atoms of carbon.

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. Match the list:

List- I	List- II
(P) SO_3	(1) 300/7 % carbon by mass
(Q) CO	(2) Empirical formula CH
(R) C_6H_6	(3) Vapour density = 40
(S) C_2H_2	(4) $40N_A$ electrons in a mole (N_A = Avogadro no.)
	(5) 50 % carbon by mole

The **CORRECT** option is :

- (A) P $\rightarrow$ 1; Q $\rightarrow$ 2; R $\rightarrow$ 1,4; S $\rightarrow$ 1,5
 (B) P $\rightarrow$ 3,4; Q $\rightarrow$ 1; R $\rightarrow$ 2,5; S $\rightarrow$ 2,5
 (C) P $\rightarrow$ 3; Q $\rightarrow$ 1; R $\rightarrow$ 2; S $\rightarrow$ 1,4
 (D) P $\rightarrow$ 2; Q $\rightarrow$ 3; R $\rightarrow$ 1; S $\rightarrow$ 4

Space for Rough Work

8. List-I

- (P) 0.5 mole of $\text{SO}_2(\text{g})$
 (Q) 1 gm molecule of $\text{O}_2(\text{g})$
 (R) 44 gm of CO_2 gas
 (S) 0.5 mole of $\text{O}_3(\text{g})$

List-II

- (1) Occupy 11.2 L at 1 atm and 273 K
 (2) Total no. of atoms = $1.5 \times N_A$
 (3) $22N_A$ electrons
 (4) weighs 32 gm
 (5) Total no. of protons = $16N_A$
 (6) Occupy 22.7 L at STP

The **CORRECT** option is :

- (A) $P \rightarrow 1$; $Q \rightarrow 2, 3, 4$; $R \rightarrow 4$; $S \rightarrow 1$
 (B) $P \rightarrow 4$; $Q \rightarrow 4, 5, 6$; $R \rightarrow 3$; $S \rightarrow 1, 2$
 (C) $P \rightarrow 2$; $Q \rightarrow 1, 2, 5$; $R \rightarrow 4$; $S \rightarrow 1$
 (D) $P \rightarrow 3$; $Q \rightarrow 1, 2, 5$; $R \rightarrow 5$; $S \rightarrow 1, 5$

Space for Rough Work

9. List- I

- (P) $\text{PCl}_5(\text{g})$
 (Q) C_6H_6
 (R) SO_3
 (S) SO_2Cl_2

List- II

- (1) polar species
 (2) involves d-orbital in hybridisation
 (3) species has atleast 4 atoms in one plane
 (4) species has atleast one $\text{p}_\pi\text{-d}_\pi$ type of π -bond
 (5) non-planar species
 (6) species has all equal bond length

The correct option is

- (A) $\text{P} \rightarrow 2,3,5$; $\text{Q} \rightarrow 1$; $\text{R} \rightarrow 3,4,6$; $\text{S} \rightarrow 1,4,5$
 (B) $\text{P} \rightarrow 1,2,3$; $\text{Q} \rightarrow 1$; $\text{R} \rightarrow 3,4,6$; $\text{S} \rightarrow 1,4,5$
 (C) $\text{P} \rightarrow 2,3,5$; $\text{Q} \rightarrow 3$; $\text{R} \rightarrow 3,4,6$; $\text{S} \rightarrow 1,4,5$
 (D) $\text{P} \rightarrow 1,2,3$; $\text{Q} \rightarrow 1,3$; $\text{R} \rightarrow 1,2,4$; $\text{S} \rightarrow 1,4,5$

Space for Rough Work

10.	List- I (Molecule)	List- II (Hybridisation of atoms present in molecules)
(P)	XeO_2F_2	(1) sp^3d^2
(Q)	BeCl_2	(2) sp^3
(R)	$\text{PCl}_5(\text{s})$	(3) sp^2
(S)	BO_3^{3-}	(4) sp
		(5) sp^3d

The correct option is

- (A) $\text{P} \rightarrow 5$; $\text{Q} \rightarrow 4$; $\text{R} \rightarrow 1,2$; $\text{S} \rightarrow 3$
 (B) $\text{P} \rightarrow 5$; $\text{Q} \rightarrow 4$; $\text{R} \rightarrow 5$; $\text{S} \rightarrow 2,3$
 (C) $\text{P} \rightarrow 5$; $\text{Q} \rightarrow 4$; $\text{R} \rightarrow 5$; $\text{S} \rightarrow 3$
 (D) $\text{P} \rightarrow 1$; $\text{Q} \rightarrow 4$; $\text{R} \rightarrow 1,2$; $\text{S} \rightarrow 3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

- Total number of species which are planar and polar.
XeF₄, XeF₂, XeF₃⁺, SO₂, H₂O, XeF₅⁻
- If the bond angle of anionic part of N₂O₅(s) is 'P' and maximum bond angle of XeO₃F₂ is 'Q'. Then find the value of (P + Q).
- How many following specie(s) have all X-X' bond lengths are same (X and X' are two different elements).
C₆H₆, ClO₄⁻, SiO₂, CO₃²⁻, HSO₄⁻, S₂O₆²⁻
- How many pair/s are present below, in which first species has lower value of electron affinity than second species:
(i) N and O (ii) O⁺ and S⁺ (iii) F⁺ and Cl⁺ (iv) I and I⁻
(v) Li⁺ and Li⁻ (vi) O and S

Space for Rough Work

5. 50 gram $\text{SO}_2(\text{g})$ allowed to react with 12 gram $\text{O}_2(\text{g})$ to form $\text{SO}_3(\text{g})$. Find amount of $\text{SO}_3(\text{g})$ (in gram) obtained at end of reaction if % yield of $\text{SO}_3(\text{g})$ is 10%.
6. A sample of carbon contains 50% C-12, 20% C-13 and 30% C-14 atoms. The average atomic mass of carbon in this sample is :
7. 20 ml of a mixture of C_2H_4 and C_2H_6 gases is burnt completely in excess of O_2 . The volume of CO_2 gas formed is :
8. To draw a line from graphite pencil requires $2.4 \mu\text{g}$ of carbon, then calculate number of moles of carbon atoms require to draw 10^7 such lines.

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following is/are correct ?

(A) $\log_2 3 > \log_3 11$

(B) $\prod_{r=2}^{1023} \log_r (r+1) = 10$

(C) If $\log_8 \left(\frac{8}{x^2} \right) = 3(\log_8 x)^2$ then sum of the solutions is $\frac{17}{8}$

(D) $\log_8 1 + \log_8 2 + \log_8 3 = \log_8 (1 + 2 + 3)$

2. If $a^{\log_5 7} = 25$ then value of $a^{(\log_5 7)^2}$ is perfect square of N then N is

(A) 7

(B) 25

(C) $\log_2 128$

(D) $2^{\log_{16} (2401)}$

Space for Rough Work

3. If characteristic of $\log_N m$ is 2 where $N = 5$ and set of all possible values of $m \in [a, b)$, then value of $\log_a b$ is
- (A) $\frac{2}{3}$ (B) $\frac{3}{2}$
- (C) rational number (D) irrational number
4. Let $T_n = (an^2 + bn + c)2^{(an^2 + bn)}$ be the n^{th} term of the series $1 + 2.2^1 + 4.2^3 + 7.2^6 + \dots$, then
- (A) $a + b = 0$ (B) $a - b - c = 0$ (C) $a^{2b} = 2$ (D) $2b + c = 0$
5. If $\left(\frac{x-1}{x}\right) + \left(\frac{x-2}{x}\right) + \left(\frac{x-3}{x}\right) + \dots$ upto $(x-1)$ terms = 3, where $x \in \mathbb{N}$ then value of x is greater than
- (A) 3 (B) 5 (C) 7 (D) 9
6. Let $S_n = \sum_{r=1}^n T_r$, where $T_r = \frac{r+2}{r(r+1)2^{r+1}}$, then
- (A) $S_\infty = 1$ (B) $S_\infty = \frac{1}{2}$
- (C) $|\log_2(1 - 2S_{15})| = 15$ (D) $|\log_2(1 - 2S_{15})| = 19$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

- | | |
|---|----------------------|
| (P) The value of $x = \log_2 \left(\log_9 \sqrt{6 + \sqrt{6 + \sqrt{6 + \dots \infty}}} \right)$, is | (1) Prime number |
| (Q) The value of $\log_{\sqrt{\frac{15}{8}}} \left(\log_{\sqrt{3}} \left(3^{\frac{1}{2}} \cdot 3^{\frac{1}{4}} \cdot 3^{\frac{1}{8}} \cdot 3^{\frac{1}{16}} \right) \right)$, is | (2) Natural number |
| (R) The value of $\left(\frac{1}{\log_5 3} + \frac{1}{\log_6 3} - \frac{1}{\log_{10} 3} \right)$ is | (3) Integer |
| (S) The smallest integer greater than $\left(\frac{1}{\log_3 \pi} + \frac{1}{\log_4 \pi} \right)$ is | (4) Composite number |
| (A) P $\rightarrow$ 3; Q $\rightarrow$ 1,2,3; R $\rightarrow$ 2,3; S $\rightarrow$ 1,2,3 | |
| (B) P $\rightarrow$ 2; Q $\rightarrow$ 3,2; R $\rightarrow$ 4; S $\rightarrow$ 3,4 | |
| (C) P $\rightarrow$ 3; Q $\rightarrow$ 2,3; R $\rightarrow$ 1,2,3; S $\rightarrow$ 1,2,4 | |
| (D) P $\rightarrow$ 2; Q $\rightarrow$ 3,4; R $\rightarrow$ 1,4; S $\rightarrow$ 3 | |

Space for Rough Work

8. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

(P) The number of solutions of the equation

(1) 4

$$\cos 6x + \tan^2 x + \cos 6x \cdot \tan^2 x = 1$$

(2) 3

in the interval $[0, 2\pi]$ is

(3) 7

(Q) The total number of solutions of

(4) 0

$$\tan x + \cot x = 2 \operatorname{cosec} x \text{ in } [-2\pi, 2\pi]$$

(R) The number of solution of the equation

$$6 \cos 2\theta + 2 \cos^2 \frac{\theta}{2} + 2 \sin^2 \theta = 0 \text{ in } (-\pi, \pi) \text{ is}$$

(S) The number of solutions of the equation

$$\sin^3 x \cos x + \sin^2 x \cos^2 x + \sin x \cos^3 x = 1$$

in the interval $[0, 2\pi]$ is

(A) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

(B) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

(C) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 1; S \rightarrow 4$

(D) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

Space for Rough Work

9. Match List-I with List-II and select the correct answer using the code given below the list.

List-I	List-II
(P) The number of solutions of $3[\sin x + \cos x] - 2[\sin^3 x + \cos^3 x] = 8$ is	(1) 4 (2) 5
(Q) The number of solutions of $\sin^4 x - \cos^2 x \sin x$ $+ 2\sin^2 x + \sin x = 0$ in $x \in [0, 3\pi]$ is	(3) 0 (4) 2
(R) The number of ordered pairs which satisfy the equation $x^2 + 2x\sin(xy) + 1 = 0$ is (where $y \in [0, 2\pi]$)	
(S) The number of values of θ which satisfy the equation $\sin 3\theta - \sin \theta = 4\cos^2 \theta - 2 \forall \theta \in [0, 2\pi]$ is	
(A) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$	(B) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$
(C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$	(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$

Space for Rough Work

10. Match List-I with List-II and select the correct answer using the code given below the list.

List-I	List-II
(P) Maximum possible value of the expression $\sqrt{\sin^4 x + 4 \cos^2 x} - \sqrt{\cos^4 x + 4 \sin^2 x}$	(1) 4 (2) 3
(Q) Maximum possible value of the expression $(\cos A - \cos B)^2 + (\sin A - \sin B)^2$	(3) 1 (4) 0
(R) Maximum possible integral value of $\sqrt{2}[\sin A + \cos A]$ is	(5) 2
(S) if $\cos \alpha \neq 1$ and $\sec(\theta + \alpha) + \sec(\theta - \alpha) = 2 \sec \theta$ then maximum possible value of $\cos \alpha$ is	
(A) $P \rightarrow 4; Q \rightarrow 5; R \rightarrow 1; S \rightarrow 3$	(B) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$
(C) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 4$	(D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If $\log x$, $\log \sqrt{6-2x}$, $\log(x-1)$ are in A.P. then value of $\frac{x}{3}$ is
2. Sum of maximum and minimum value of $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$ is
3. The least integral value of ' λ ' for which the expression $(\lambda x^2 - 4x + 3\lambda + 1)$ is positive for all $x \in \mathbb{R}$, is

Space for Rough Work

-
4. If α, β, γ are roots of $x^3 - 5x^2 + 3 = 0$ then value of $|(\alpha - 2)(\beta - 2)(\gamma - 2)|$ is
5. Let $p = 1 + \frac{1}{2} + \frac{1}{2^2} + \dots \infty$ and q be the 20th term of 0, 1, 2, 3, 4, 5 ... then $\frac{q}{p}$ is
6. Number of integral roots of the equation $8x^{100} - 20x^4 + 3x^2 - 91 = 0$
7. Minimum possible integral value of k such that equation $2^{2x} - 2(k - 1)2^x + k = 0$ has one root less than 1 and other root is greater than 1.
8. If $\frac{\sec 8A + \sec 4A - 2}{\sec 8A - \sec 4A} = \frac{\sin nA}{\sin mA}$, then $\frac{m}{n}$ is equal to
-

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Matching Lists Type (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator